

CS472 Software Testing and Continuous Integration Lab Report

Peter Nguyen

Introduction

The following report follows my work across the three phases, each covering test coverage, test driven development, and continuous integration. Each phase followed a similar workflow of synchronizing the work, defining the work as an issue with an acceptance criteria, and then implementing a focused branch.

Fork Repository: <https://github.com/Peekans/Group-2>

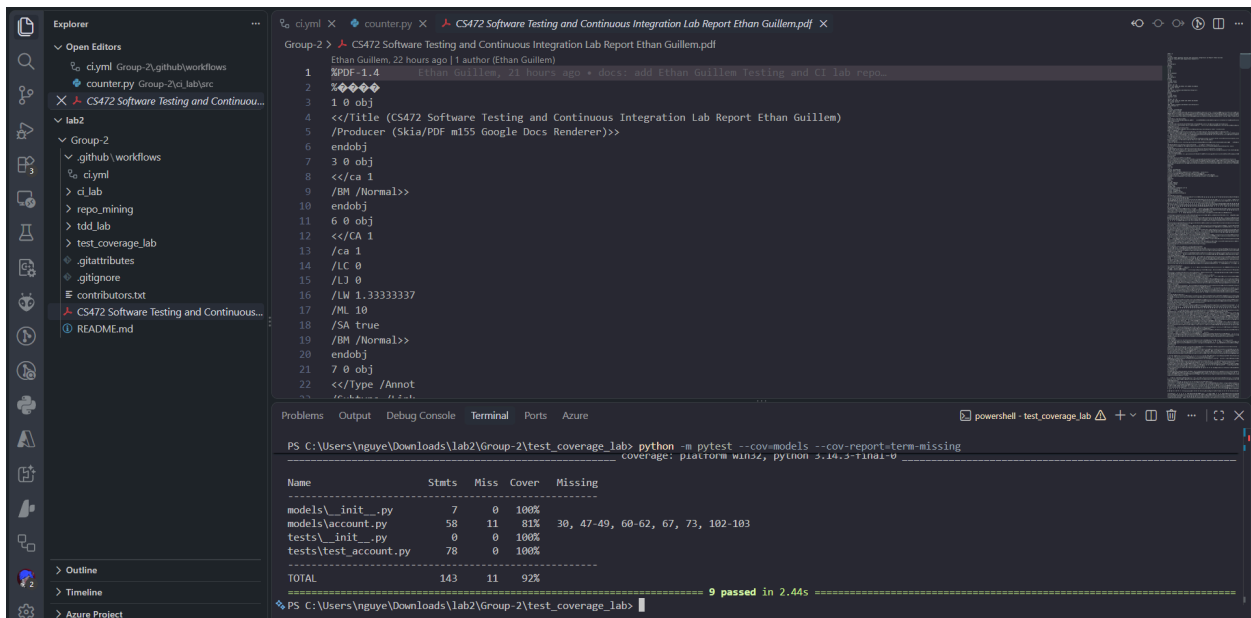
Team upstream repository: <https://github.com/kenner1-unlv/Group-2>

My assigned reviewer: Michael Podolsky. Github: MikePodo

Person I was responsible for reviewing: Jonathan Johnson. Github: <https://github.com/johns216>

Phase I

Setup evidence:



The screenshot shows a VS Code editor with a PDF file open, titled "CS472 Software Testing and Continuous Integration Lab Report Ethan Guillem.pdf". The PDF content is visible, showing a title, author, and a table of contents. Below the editor, a terminal window displays the output of a Python test command. The terminal shows the command: `PS C:\Users\nguye\Downloads\lab2\Group-2\test_coverage_lab> python -m pytest --cov=models --cov-report=term-missing`. The output shows a table of test results with columns: Name, Stmts, Miss, Cover, and Missing. The table lists several test files and their coverage percentages. The total coverage is 92%.

Name	Stmts	Miss	Cover	Missing
models__init__.py	7	0	100%	
models\account.py	58	11	81%	30, 47-49, 60-62, 67, 73, 102-103
tests__init__.py	0	0	100%	
tests\test_account.py	78	0	100%	
TOTAL	143	11	92%	

9 passed in 2.44s

Link to issue: <https://github.com/kenner1-unlv/Group-2/issues/76>

Link to PR: <https://github.com/kenner1-unlv/Group-2/pull/77>

Coverage Report:

```
PS C:\Users\nguye\Downloads\lab2\Group-2\test_coverage_lab> python -m pytest --cov=models --cov-report=term-missing
===== test session starts =====
platform win32 -- Python 3.14.3, pytest-8.4.2, pluggy-1.6.0 -- C:\Python314\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\nguye\Downloads\lab2\Group-2\test_coverage_lab
configfile: pytest.ini
plugins: cov-7.1.0
collected 6 items

tests/test_account.py::test_account_role_assignment PASSED [ 16%]
tests/test_account.py::test_invalid_role_assignment PASSED [ 33%]
tests/test_account.py::test_password_hashing PASSED [ 50%]
tests/test_account.py::test_account_deactivation_and_reactivation PASSED [ 66%]
tests/test_account.py::test_valid_withdrawal PASSED [ 83%]
tests/test_account.py::test_invalid_email_input PASSED [100%]

===== tests coverage =====
coverage: platform win32, python 3.14.3-final-0
-----
Name                               Stmts  Miss  Cover   Missing
-----
models\__init__.py                   7      0  100%
models\account.py                   58     16   72%    30, 34, 53-56, 60-62, 66-68, 73, 75, 102-103
tests\__init__.py                     0      0  100%
tests\test_account.py               61      0  100%
-----
TOTAL                               126     16   87%

===== 6 passed in 2.12s =====
```

Explanation: I did pool item 2, and to verify the item, I tested invalid email inputs against `validate_email()`, in function `test_invalid_email_input` in `test_account.py`. There was a list of string literals containing illegal emails such as emails using white space, one with no domains, no `@` ... etc, and ensured that the `DataValidationError` was raised, passed locally.

Link to teammate's

PR: <https://github.com/kenner1-unlv/Group-2/pull/77#pullrequestreview-5239503863>

Link to my review

comment: <https://github.com/kenner1-unlv/Group-2/pull/94#issuecomment-5736573874>

Short summary of how I received feedback:

Mike corrected my error in creating a concatenated single element of invalid emails, rather than single individual invalid emails, as I had forgotten to include commas in between each string. This was immediately fixed, followed with a comment on both ends approving for the merge.

Link to approved PR:

<https://github.com/kenner1-unlv/Group-2/pull/77#pullrequestreview-5252942174>

Phase II

Setup evidence:

The screenshot shows a VS Code editor with a file explorer on the left and a code editor on the right. The file explorer shows a project structure with a file named `test_counter.py`. The code editor shows the following Python code:

```
77 class TestCounterEndpoints:
98     def test_reset_all_counters(self, client):
107
108         assert result.status_code == status.HTTP_200_OK
109         assert result.get_json() == {'message': 'All counters reset'}
110         assert COUNTERS == {'alpha': 0, 'beta': 0}
111
112     def test_retrieve_nonexistent_counter(self, client):
113         """It should return 404 when retrieving a counter that does not exist"""
114         result = client.get('/counters/nonexistent')
115
116         assert result.status_code == status.HTTP_404_NOT_FOUND
```

The output window shows the results of running the tests:

```
PS C:\Users\nguye\Downloads\lab2\Group-2\tdd_lab> python -m pytest --cov=src
configfile: pytest.ini
plugins: cov-7.1.0
collected 8 items

tests/test_counter.py::TestCounterEndpoints::test_create_counter PASSED [ 12%]
tests/test_counter.py::TestCounterEndpoints::test_delete_counter PASSED [ 25%]
tests/test_counter.py::TestCounterEndpoints::test_reject_invalid_counter_name PASSED [ 37%]
tests/test_counter.py::TestCounterEndpoints::test_list_counters PASSED [ 50%]
tests/test_counter.py::TestCounterEndpoints::test_increment_counter PASSED [ 62%]
tests/test_counter.py::TestCounterEndpoints::test_retrieve_existing_counter PASSED [ 75%]
tests/test_counter.py::TestCounterEndpoints::test_reset_all_counters PASSED [ 87%]
tests/test_counter.py::TestCounterEndpoints::test_retrieve_nonexistent_counter PASSED [100%]

===== tests coverage =====
coverage: platform win32, python 3.14.3-final-0

Name          Stmts  Miss  Cover   Missing
-----
src/_init_.py    3      0    100%
src/counter.py  38      1     97%   33
src/status.py    7      0    100%

TOTAL          48      1     98%

===== 8 passed in 0.63s =====
PS C:\Users\nguye\Downloads\lab2\Group-2\tdd_lab>
```

Team's TDD setup PR: <https://github.com/kenner1-unlv/Group-2/pull/47>

Issue: <https://github.com/kenner1-unlv/Group-2/issues/46>

Link to my TDD PR: <https://github.com/kenner1-unlv/Group-2/pull/92>

Test case I wrote: <https://github.com/kenner1-unlv/Group-2/pull/92/files>

Brief explanation:

I did pool item 3, where I had to return 404 for a non-existent counter, targeting GET/counters/<name>. Before my change the route indexed COUNTERS[name] directly with no existence check, so requesting a counter that had never been created raised an unhandled KeyError and Flask returned HTTP 500.

RED test result:

tests/test_counter.py::TestCounterEndpoints::test_retrieve_nonexistent_counter FAILED [100%]

....

```
=====
===== tests coverage
=====
=====
```

coverage:
platform win32, python 3.14.3-final-0

Name	Stmts	Miss	Cover	Missing
------	-------	------	-------	---------

src__init__.py	3	0	100%	
src\counter.py	36	1	97%	33
src\status.py	7	0	100%	

TOTAL	46	1	98%	
-------	----	---	-----	--

===== short test summary info

FAILED

tests/test_counter.py::TestCounterEndpoints::test_retrieve_nonexistent_counter -
assert 500 == 404

===== 1 failed, 7 passed in 0.53s

GREEN test result:

tests/test_counter.py::TestCounterEndpoints::test_retrieve_nonexistent_counter
PASSED [100%]

....

===== tests coverage

coverage:
platform win32, python 3.14.3-final-0

Name	Stmts	Miss	Cover	Missing
------	-------	------	-------	---------

```

-----
src\__init__.py    3    0 100%
src\counter.py    38    1 97%  33
src\status.py     7    0 100%
-----
TOTAL              48    1 98%
=====
===== 8 passed in 0.44s
=====
=====

```

Reason for no refactor:

No refactoring was necessary. The `counter_exists()` helper had already been extracted during the worked example in the team setup PR, and the correct implementation of this feature was to reuse it rather than duplicate the membership test. Introducing a new abstraction here would have added indirection without improving structure, so the minimum GREEN implementation was also the final one.

Summary of reviewer feedback and how I addressed it:

Michael brought up the fact that I used an invalid name for my test, as no matter what, I would've been able to pass since it was a false positive as I used hyphens in my variable "does-not-exist". I addressed this by promptly using a valid name, using "nonexistent" instead and getting approved afterwards.

Link to teammate's PR comment:

<https://github.com/kenner1-unlv/Group-2/pull/92#pullrequestreview-5250180190>

Link to teammate I was responsible for PR:

<https://github.com/kenner1-unlv/Group-2/pull/97>

Brief statement of evidence I examined:

For the test results, I noticed that when my teammate tested for the RED phase, it actually had all passes, which did not indicate a proper RED phase. Additionally, in their GREEN phase, it had an additional pass that was not possible without other tamperment or misaligned code, as the passed test cases went from 7 to 9.

My comment in their PR:

<https://github.com/kenner1-unlv/Group-2/pull/97#issuecomment-5737173308>

Link to approved comment for my PR:

<https://github.com/kenner1-unlv/Group-2/pull/92#pullrequestreview-5250704732>

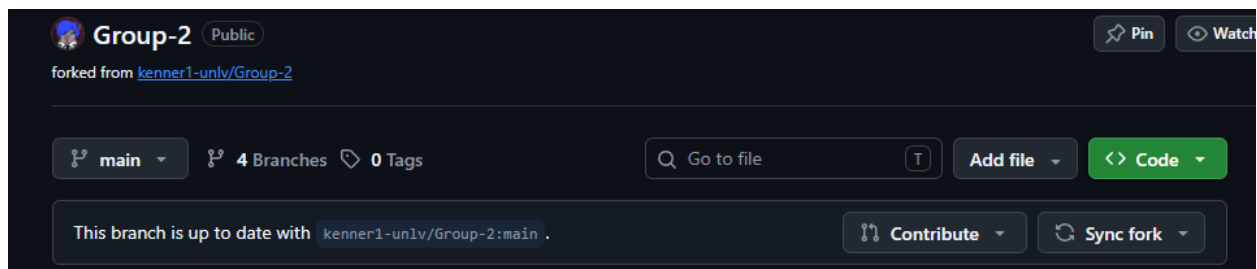
Link to my teammate's approved PR from my comment:

<https://github.com/kenner1-unlv/Group-2/pull/97#pullrequestreview-5254003387>

Phase III

Link to my fork: <https://github.com/Peekans/Group-2>

Evidence that my fork was synchronized:



Evidence that the ci_lab starter workflow ran successfully:

```
PS C:\Users\nguye\Downloads\lab2\Group-2\ci_lab> python -m pytest --cov=src --cov-report=term-missing
tests/test_counter.py::TestCounterEndpoints::test_handle_invalid_http_methods PASSED [ 50%]
tests/test_counter.py::TestCounterEndpoints::test_get_total_counters PASSED [ 54%]
tests/test_counter.py::TestCounterEndpoints::test_top_n_counters PASSED [ 59%]
tests/test_counter.py::TestCounterEndpoints::test_bottom_n_counters PASSED [ 63%]
tests/test_counter.py::TestCounterEndpoints::test_set_counter_to_value PASSED [ 68%]
tests/test_counter.py::TestCounterEndpoints::test_prevent_negative_counter_values PASSED [ 72%]
tests/test_counter.py::TestCounterEndpoints::test_reset_single_counter PASSED [ 77%]
tests/test_counter.py::TestCounterEndpoints::test_prevent_resetting_non_existent_counter PASSED [ 81%]
tests/test_counter.py::TestCounterEndpoints::test_get_total_number_of_counters PASSED [ 86%]
tests/test_counter.py::TestCounterEndpoints::test_counters_greater_than_threshold PASSED [ 90%]
tests/test_counter.py::TestCounterEndpoints::test_counters_less_than_threshold PASSED [ 95%]
tests/test_counter.py::TestCounterEndpoints::test_validate_counter_name PASSED [100%]

===== tests coverage =====
coverage: platform win32, python 3.14.3-final-0

Name           Stmts  Miss  Cover   Missing
-----
src\__init__.py    3     0   100%
src\counter.py   87     5    94%   71, 85, 96, 99-100
src\status.py     7     0   100%
TOTAL             97     5    95%

===== 22 passed in 0.66s =====
PS C:\Users\nguye\Downloads\lab2\Group-2\ci_lab>
```

Link to my

issue: <https://github.com/kenner1-unlv/Group-2/issues/93#issue-5502135400>

Link to my PR: <https://github.com/kenner1-unlv/Group-2/pull/95>

Link to workflow excerpt: <https://github.com/kenner1-unlv/Group-2/pull/95/files>

Link to failing CI log and root-cause summary:

<https://github.com/kenner1-unlv/Group-2/pull/95#issue-5502783294>

Link to fix

commit: <https://github.com/kenner1-unlv/Group-2/commit/580ad20a8c3c38cee29ab7932dbfed62c9a83a05>

Link to successful CI run:

<https://github.com/kenner1-unlv/Group-2/pull/95#issuecomment-5734256151>

And

<https://github.com/kenner1-unlv/Group-2/actions/runs/35378917305>

Link to teammate's PR I

reviewed: <https://github.com/kenner1-unlv/Group-2/pull/102>

Brief statement of evidence I examined while reviewing:

I reviewed Jonathan Johnson's PR for Enhancement 3, and I examined the ci.yml diff showing the narrowed Flake8 --select list and the added Black install and check steps. I examined both controlled failure logs. The Flake8 run listed E302, E501, W292, and W391 violations with exit code 1, and the Black run reporting two files it would reformat, along with the fix commits for each. I examined the final successful run showing Lint, Coverage, and the three-version test matrix all green. My approving comment noted that some unrelated whitespace changes were included, which will resolve once the other PRs merge.

Link to my comment:

<https://github.com/kenner1-unlv/Group-2/pull/102#pullrequestreview-5254192926>

Reflection:

The biggest lesson that I took home was that passing a test does not automatically mean it's a meaningful one. Especially in my phase 2, where Michael caught that even though the test passed, the status code was correct. It also made me realize how helpful PR's are and having peer reviews can massively increase productivity and fix issues.

I've also gotten into that habit of Continuous Integration, which I believe will greatly help me moving forward. The value isn't just that the checks exist, it's that they run on someone else's machine, on Python versions I don't have installed, before anything can be merged. Setting up my own enhancement made me realize how much a team relies on that feedback actually reaching people.

Control Failure was a brand new perspective for me, and was not what I'm used to, but breaking something on purpose taught me what's beneath the surface in a way I wouldn't have figured out otherwise.